



Republic of Rwanda
Ministry of Education



RTB | RWANDA
TVET BOARD

LINEAR DISTANCE MEASUREMENTS

LSVDM302

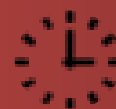
Measure linear distance

Competence

RQF Level: 3

Learning Hours

Credits: 6



60

Sector: Construction and Building Services

Trade: Land surveying

Module Type: Specific

Curriculum: CBSLSV3001 - TVET Certificate 3 in Land Surveying

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Purpose statement	This module is designed for students pursuing a level 3 TVET certificate in Land Surveying. It allows the learner to get the knowledge, skills, and attitude required to perform both direct and indirect distance measurement.					
Leaning assumed to be in place	Safety, Health and environment at workplace; Surveying tools and equipment; Surveying communication language.					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	
	• Group project and presentation	20%				
	• Individual project /Work	50%				
			Summative Assessment		50%	

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria	
1. Prepare the site	1.1.	Site reconnaissance is well conducted according to the information required.
	1.2.	Tools, materials, and equipment are well selected according to the given task.
	1.3.	Site clearing is well performed according to the work requirements.
2. Perform direct distance measurements	Distance measuring methods are properly identified according to the types of work.	
	2.1.	Tools and equipment are appropriately selected according to the accuracy required.

	2.2. Measurements and marking are well performed according to selected methods.
	2.3. Data recording and processing are well performed according to the work requirement.
3. Perform indirect distance measurements	3.1. Methods of distance measurements are properly identified according to the type of work.
	3.2. Tools and equipment are appropriately selected according to the accuracy of work.
	3.3. Data measuring, recording, and processing are well performed according to the type of work.

Learning outcomes	At the end of the module the learner will be able to: 1. Prepare the site 2. Perform direct distance measurements 3. Perform indirect distance measurements
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Learning Outcome 1: Prepare the site		Learning hours: 7
Indicative content		
* Information required in reconnaissance • Topography • Site accessibility and utility services (roads ...) * Tools and materials needed during site preparation • Tools (machete, pickaxe, etc) • Materials (PPE: helmet, gloves, etc) * Site clearing Refer to RS ISO 14015 • Remove trees, brushes, etc		
Resources required for the learning outcome		
Equipment	Compass	
Materials	Pegs, nails, paint/chalk, eraser, ground mark, pencil/pen/ marker pens, pins, field notes.	

Tools	Ranging rod, arrows, measuring tape, hammers, building line, machete pickaxe hoe.
Facilitation techniques	• Demonstration and simulation • Individual and group work • Individualized • Trainer guided • Group discussion
Formative assessment methods	• Written assessment • Oral assessment • Performance assessment

Learning Outcome 2: Perform Direct distance Measurements	Learning hours: 25
Indicative content	
* Methods used indirect measurements (Refer to ASTM F12203-134) • Surveying working principles (working from the whole to the part, consistency ...) • Methods of direct distance measurements (pacing, taping/ chaining, measuring wheel) * Tools and equipment used in direct distance measurements • Tools (ranging rod, arrows, chain, tape, staffs, plumb bob) • Materials (pegs, building lines ...) • Equipment (EDM, laser meter, etc.) * Linear Measurements and marking • Establishing baseline • Ranging out survey line (direct ranging and indirect or optical ranging) • Offsets (Perpendicular and inclined) • Survey station (main station and subsidiary or tie stations) 2.4. Data measuring, recording, and processing • Booking format (electronic, manual field book) • Data recording techniques (tabular, sketches, etc) • Data interpretation/Analysis (calculation/computations)	

• Correction of errors in taping (length, slope, sag, tension,etc)	
Resources required for the indicative content	
Equipment	EDM, Computer, laser meter, whiteboard, projector.
Materials	Notebook, pen, marker pen, pieces of chalk.
Tools	Ranging rod, arrows, chain, tape, staffs, plumb bob , tape measure
Facilitation techniques	• Demonstration and simulation • Individual and group work • Practical exercise • Individualized • Trainer guided • Group discussion
Formative assessment methods	• Written assessment • Oral assessment • Performance assessment • Product based assessment

Learning Outcome 3: Perform indirect distance measurements	Learning hours: 28
Indicative content	
* Identification of methods used in indirect measurements • Measurement by optical methods (tachometry survey) * Tools and instruments used in indirect measurements • Tools (tape measure, levelling staff, etc.) • Instruments (levels, theodolite, etc.) * Data measuring, recording, and processing	

• Booking format (electronic, manual field book) • Procedure of data collection • Recording techniques (tabular, sketches, notes) • Data interpretation (Calculation/Computations of data) • Error adjustment	
Resources required for the indicative content	
Equipment	Levels, theodolite, whiteboard, projector, computer.
Materials	Notebook, pen, pegs, marker pen, pieces of chalk.
Tools	Tape measure, plumb bob, ranging rod, arrow and pins, nails.
Facilitation techniques	• Demonstration and simulation • Individual and group work • Practical exercise • Individualized • Trainer guided • Group discussion
Formative assessment methods	• Written assessment • Oral assessment • Performance assessment • Product based assessment

Integrated/Summative assessment

Integrated situation

The Kigali City in collaboration with an American fund project has initiated a project of city development. They want to construct standardized entertainment facilities (football, volleyball grounds) in Kicukiro/Gahanga sector to overcome gaps in the sports sector. Their measurements are 100m of long and 64m in width, 24m and 18m respectively. Tender notice has been given and JACOBS's company Ltd has become a successful bidder. As a land surveyor of JACOBS company Ltd, you are requested to carry out the measurements of these facilities on the ground by using levelling instrument for the football ground and taping for the volleyball ground. Drawings and instruments are provided by the company. The work should be

completed in 6 hours.

Resources

Equipment	level and its accessories.
Tools	Machete, tape measure, ranging rod, staff reading, hammer, plumb.
Materials	Pegs, nails, building lines, notebook, pen/pencil, paint/chalk, eraser.

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
1. Prepare the site (15%)	1.2. Tools, materials, and equipment are selected.	1. Tools are selected.			4
		2. Materials are selected.			5
		3. Equipment is selected.			6
2. Perform Direct distance Measurements (45%)	2.1. Measuring methods applied.	1. Taping is applied.			6
	2.3. Measurements and marking are performed.	1. Distance is measured.			8
		2. Points are marked.			6
		3.Points are well aligned.			8
	2.4. Data recording and processing are well performed	1. Distances are recorded.			9
		2. Dimensions are respected.			8
3. Perform indirect distance measurements (40%)	3.2. Field procedures are respected.	1. level is well set up.			5
		2. Centering is well done.			5
		3. Focusing is well done.			3
	3.3. Data recording and processing are well performed	1. Upper reading is well recorded.			5
		2. Middle reading is well recorded.			7
		3. Lower reading is well recorded.			7
		4. Horizontal distance is accurately computed.			8
Total marks		100			

Percentage Weightage	100%
Minimum Passing line % (Aggregate): 70%	

References:

1. AMIRAJ. (2009). Tacheometric surveyin. New York.
2. Basak, N. Surveying and Leveling.
3. Breach, W. (2007). Engineering Surveying Six Edition. Oxford.
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5. UNESCO, V. (December 2008). National Diploma in building Technology.
6. WOLF, C. (2012). Elementary Surveying Thirteenth Edition. New Jersey.
7. ASTM F2203-13 Standard Test Method For Linear Measurement Using Precision Steel Rule
8. RS ISO 14015: Environmental management systems — Environment Assessment of Sites and Organizations